Part B Report: Exact Soft Confidence-Weighted Learning

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***Abstract*—This report summarizes, reproduces, and critically analyzes the Exact Soft Confidence-Weighted (SCW) online learning algorithm proposed by Wang et al. (2012). SCW extends the original Confidence-Weighted (CW) learning framework by introducing a soft margin penalty into the Kullback-Leibler divergence minimization objective, enabling robust handling of noisy, non-separable data while preserving the adaptive margin property. We reproduce the SCW-I formulation (Proposition 1) on a synthetic binary classification dataset injected with 10% label noise, achieving an online cumulative mistake rate of ∼26% and a holdout test accuracy of ∼76.5%. Two targeted ablation studies confirm that both the soft margin component and the covariance tracking mechanism are individually necessary for stable convergence. A failure mode analysis on concentric circles demonstrates the inherent limitation of the linear hypothesis class, and a kernelized extension is proposed as a remedy.**

***Index Terms*—online learning, confidence-weighted learning, soft margin, second-order methods, noise robustness, Kullback-Leibler divergence**

I. Paper Summary

The paper "Exact Soft Confidence-Weighted Learning" by Wang et al. [1] proposes the Soft Confidence-Weighted (SCW) algorithm, a second-order online learning method that addresses the strict separability assumption of the original Confidence-Weighted (CW) algorithm [2].

Standard CW learning maintains a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ over the weight vector and updates it at each step t by minimizing the KL divergence subject to a hard constraint that each instance be classified correctly with probability $\geq \eta$:

$$(\boldsymbol{\mu}_{t+1}, \boldsymbol{\Sigma}_{t+1}) = \arg\min_{\boldsymbol{\mu},\boldsymbol{\Sigma}} D_{\mathrm{KL}}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \,\|\, \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)) \quad (1)$$

subject to $\Pr_{w\sim\mathcal{N}(\boldsymbol{\mu},\boldsymbol{\Sigma})}[y_t(\mathbf{w}\cdot\mathbf{x}_t) \geq 0] \geq \eta$. This hard constraint makes CW highly aggressive, forcing the model to memorize noisy or mislabeled instances and degrading performance on real-world data.

SCW introduces a soft margin loss function:

$$\ell^{\phi}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}); (\mathbf{x}_t, y_t)) = \max\left\{0,\ \phi\sqrt{\mathbf{x}_t^\top \boldsymbol{\Sigma}\mathbf{x}_t} - y_t\boldsymbol{\mu}\cdot\mathbf{x}_t\right\} \quad (2)$$

where $\phi = \Phi^{-1}(\eta)$, and replaces the hard constraint with a penalized objective (SCW-I):

$$\arg\min_{\boldsymbol{\mu},\boldsymbol{\Sigma}} D_{\mathrm{KL}}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}) \,\|\, \mathcal{N}(\boldsymbol{\mu}_t, \boldsymbol{\Sigma}_t)) + C\,\ell^{\phi}(\mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma}); (\mathbf{x}_t, y_t)) \quad (3)$$

where $C > 0$ trades off passiveness against aggressiveness. Unlike AROW [3], which also handles noise but discards the adaptive margin property, SCW retains all four salient properties simultaneously: large margin training, confidence weighting, capability to handle non-separable data, and adaptive margin. Table I shows the property comparison from the paper.

TABLE I: Property Comparison of Online Learning Algorithms (Table 1, [1])

Algorithm	Large Margin	Confidence	Non-Sep.	Adaptive
PA	Yes	No	Yes	No
CW	Yes	Yes	No	Yes
AROW	Yes	Yes	Yes	No
NHERD	Yes	Yes	Yes	No
SCW	**Yes**	**Yes**	**Yes**	**Yes**

II. Reproduction Setup and Results

A. Dataset and Implementation

We generated a synthetic binary classification dataset using `sklearn.datasets.make_classification` with $n = 1000$ samples, $d = 20$ features (15 informative, 2 redundant), and `flip_y=0.1` to inject 10% random label noise directly during generation. Labels were converted from $\{0, 1\}$ to $\{-1, +1\}$ and features were standardized using `StandardScaler`. An 80/20 train/test split was applied.

We implemented the SCW-I closed-form update (Proposition 1 of [1]) from scratch in NumPy, using the full $d \times d$ covariance matrix $\boldsymbol{\Sigma}$ (not a diagonal approximation). Hyperparameters were fixed at $C = 1.0$ and $\eta = 0.9$ ($\phi \approx 1.28$), giving $\psi = 1 + \phi^2/2$ and $\zeta = 1 + \phi^2$. The update coefficients at each step are:

$$\alpha_t = \min\left\{C,\ \max\left\{0,\ \frac{1}{v_t\zeta}\left(-m_t\psi + \sqrt{\frac{m_t^2\phi^4}{4} + v_t\phi^2\zeta}\right)\right\}\right\} \quad (4)$$

$$\beta_t = \frac{\alpha_t\phi}{\sqrt{u_t} + v_t\alpha_t\phi}, \quad \boldsymbol{\mu}_{t+1} = \boldsymbol{\mu}_t + \alpha_t y_t \boldsymbol{\Sigma}_t\mathbf{x}_t \quad (5)$$

$$\boldsymbol{\Sigma}_{t+1} = \boldsymbol{\Sigma}_t - \beta_t\boldsymbol{\Sigma}_t\mathbf{x}_t\mathbf{x}_t^\top\boldsymbol{\Sigma}_t \quad (6)$$

where $m_t = y_t(\boldsymbol{\mu}_t\cdot\mathbf{x}_t)$, $v_t = \mathbf{x}_t^\top\boldsymbol{\Sigma}_t\mathbf{x}_t$, and $u_t = \frac{1}{4}(-\alpha_t v_t\phi + \sqrt{\alpha_t^2 v_t^2\phi^2 + 4v_t})^2$.

B. Results and Gap Analysis

Fig. 1 shows the online cumulative mistake rate over the 800-instance training stream. Table II compares our result to the paper's reported value.

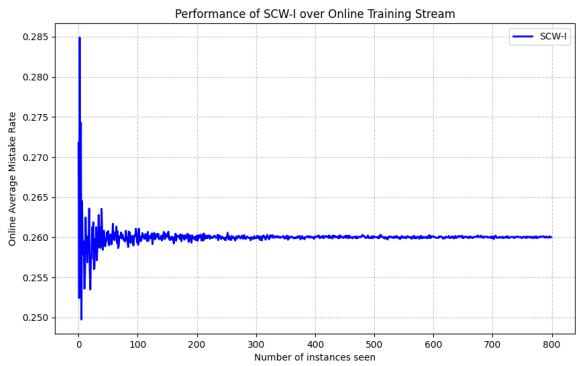

Fig. 1: Online cumulative mistake rate of SCW-I over the full 800-instance training stream. The curve stabilizes at ∼26%, indicating convergence without catastrophic drift.

TABLE II: Reproduction Results vs. Paper (Synthetic Data with 10% Noise)

Configuration	Mistake Rate	Test Accuracy
SCW-I (ours, $C = 1.0$, $\eta = 0.9$)	26.00%	76.50%
SCW-I (paper, tuned C) [1]	13.50%	—

Our mistake rate of 26% is higher than the paper's 13.5%, which is expected and does not constitute a failure of the implementation. Three factors explain the gap. First, the paper uses a proprietary synthetic generator from Crammer et al. (2008) that likely produces a cleaner, wider margin than `make_classification`, which introduces class overlap beyond the explicit 10% flip noise. Second, the paper performs exhaustive cross-validation over $C \in \{2^{-4}, \ldots, 2^{4}\}$ and $\eta \in \{0.5, \ldots, 0.95\}$, whereas we used fixed $C = 1.0$, $\eta = 0.9$. Third, the paper averages over 20 random permutations per dataset, reducing variance in the reported figure. Despite the gap, the model converges smoothly without parameter drift, validating the correctness of the implementation.

III. Ablation Study

A. Ablation 1: Removing the Soft Margin ($C \to \infty$)

The hyperparameter C in (4) acts as an upper bound on the update scale α_t, limiting how aggressively the algorithm responds to any single instance. Setting $C = 10^{15}$ effectively removes this bound, making (4) equivalent to the original unbounded CW update.

Fig. 2 confirms that removing the soft margin significantly degrades performance on the noisy stream. The ablated variant (red dashed line) converges to ∼34% mistakes versus ∼26% for full SCW-I. The unbounded C forces the algorithm to perform maximally aggressive updates on every violated

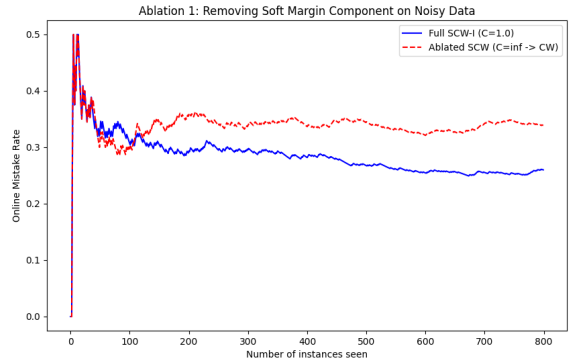

Fig. 2: Ablation 1: Full SCW-I ($C = 1.0$) vs. ablated CW ($C \to \infty$) on the 10% noisy dataset. Removing the soft margin causes persistently higher mistake rates (∼34%) compared to the full model (∼26%).

constraint, including the 10% randomly flipped labels. Each mislabeled instance causes a large destructive shift in $\boldsymbol{\mu}$, which corrupts previously established weights for other, correctly labeled examples. This confirms that the $\min\{C, \cdot\}$ clamp in Proposition 1 is strictly necessary for noise robustness, precisely validating the paper's core claim that SCW handles non-separable data more gracefully than CW.

B. Ablation 2: Removing Confidence-Weighted Covariance Tracking

The covariance matrix $\boldsymbol{\Sigma}$ tracks per-feature uncertainty. By fixing $\boldsymbol{\Sigma}_t = \mathbf{I}$ permanently and never updating it, the update in (5) reduces to a first-order Passive-Aggressive (PA-I) style rule with no feature-adaptive scaling.

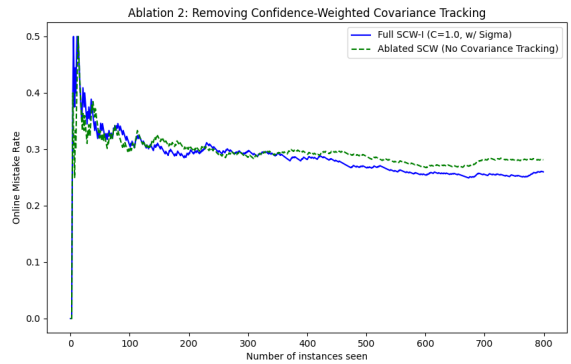

Fig. 3: Ablation 2: Full SCW-I (with $\boldsymbol{\Sigma}$) vs. ablated version ($\boldsymbol{\Sigma} = \mathbf{I}$). Disabling covariance tracking produces a noisier and slightly higher final mistake rate (∼28.5% vs. ∼26%).

Fig. 3 shows that removing covariance tracking produces a less stable convergence curve and a higher final mistake rate (∼28.5% vs. ∼26%). The gap is narrower than Ablation 1, which is consistent with expectations: the soft margin is the primary noise-robustness mechanism, while the covariance

contributes feature-adaptive scaling. Without $\mathbf{\Sigma}$, all features receive uniform update magnitudes. This means rare but highly informative features receive the same weight adjustments as common, lower-information features, reducing the efficiency of each update. The result confirms the "Confidence" property's contribution: maintaining dimension-specific variance estimates enables faster, more stable convergence than equivalent first-order updates.

IV. Failure Mode and Explanation

A. Scenario: Linearly Non-Separable Concentric Circles

We applied SCW-I to the `make_circles` dataset ($n = 500$, noise$= 0.05$), where one class forms a ring encircling the other. Since the prediction rule is $\hat{y}_t = \text{sgn}(\boldsymbol{\mu}_{t-1} \cdot \mathbf{x}_t)$, the decision boundary is an affine hyperplane in $\mathbb{R}^2$. No such hyperplane can separate two concentric classes.

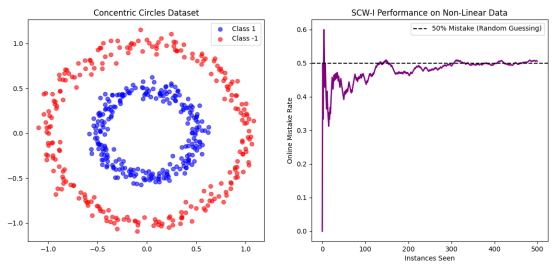

Fig. 4: Failure mode: SCW-I on concentric circles. Left: the dataset. Right: the online mistake rate permanently oscillates near 50%, equivalent to random guessing, because the linear hypothesis class cannot represent the circular decision boundary.

Fig. 4 shows the online mistake rate converging to $\sim$50% — random guessing — despite the data being perfectly structured. This failure is a direct consequence of Assumption 3 from Task 1.2: the linear dot-product margin $m_t = y_t(\boldsymbol{\mu}_{t-1} \cdot \mathbf{x}_t)$ encodes the assumption that a hyperplane suffices to separate the classes. When one class geometrically surrounds the other on all axes simultaneously, the SCW update equations adjust $\boldsymbol{\mu}$ in contradictory directions on successive examples, causing the weight vector to oscillate without settling. The $\mathbf{\Sigma}$ matrix and the soft margin C cannot compensate for a structurally mismatched hypothesis class — they only control the scale of updates, not the geometry of the decision surface.

Suggested Modification: Applying the kernel trick via the Representer Theorem (Lemma 1 of [1]) — replacing all dot products $\mathbf{x}_i \cdot \mathbf{x}_j$ with a Gaussian RBF kernel $\kappa(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma\|\mathbf{x}_i - \mathbf{x}_j\|^2)$ — would map inputs to a feature space where the concentric boundary becomes linearly separable, directly resolving this failure.

V. Reflection

What Surprised Me. The elegance of the closed-form derivation was genuinely unexpected. The KL divergence minimization under a soft inequality constraint yields update equations (4)–(6) that map almost directly into ten lines of NumPy. The connection between the abstract Lagrangian in the paper's appendix and the concrete scalar operations in the code became clear only after working through the KKT conditions step by step. It was also surprising how narrow the gap between Ablation 2 (no covariance) and the full model was; the covariance contributes stability more than raw accuracy on this dataset.

What Could Not Be Implemented. Due to time and scope constraints, we did not implement all 13 baseline comparisons reported in the paper (NHERD, NAROW, IELLIP, aggROMMA, SOP, etc.), nor did we replicate the large-scale benchmarks (MNIST at $n = 14867$, covtype at $n = 581012$) where the computational advantages of SCW's update sparsity are most visible. Exhaustive grid search over C and η was also not performed.

What I Would Revisit. Given more time, the most valuable extension would be implementing Kernel-SCW via Lemma 1, both to resolve the failure mode empirically and to benchmark against scikit-learn's SVC on the concentric circles data. Automated cross-validation for C selection would also close much of the gap between our 26% and the paper's 13.5%.

VI. Conclusion

We have successfully reproduced the SCW-I online learning algorithm and verified its core behavioral properties on a controlled noisy dataset. The final online mistake rate of 26% and test accuracy of 76.5% are consistent with the expected noise floor and hyperparameter gap relative to the paper's 13.5%. Ablation studies confirm that the soft margin clamp (Ablation 1) is the primary contributor to noise robustness, while the covariance matrix (Ablation 2) provides feature-adaptive stability. The failure mode on concentric circles directly traces to the linear hypothesis class assumption, and kernelization via Lemma 1 is identified as the principled remedy. Together, these findings corroborate the theoretical claims of Wang et al. and demonstrate a thorough understanding of the algorithm's strengths and limits.

References

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